

Enhancing Person Detection in Blurred Environments Using Fine-Tuned YOLOv8

Computer Vision Final Project

This notebook demonstrates the complete workflow for training, validating, and evaluating a YOLOv8 model for robust person detection in blurred environments.

```
In [9]: from google.colab import drive  
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call `drive.mount("/content/drive", force_remount=True)`.

```
In [10]: from ultralytics import YOLO  
import pandas as pd  
import matplotlib.pyplot as plt  
from PIL import Image  
import cv2  
import os
```

Data Preprocessing

A custom blurred-person dataset was created for this project. Since the dataset was specifically developed for YOLOv8 training, the images and annotations were already organized in the required YOLO format.

Therefore, extensive preprocessing was not required. The preprocessing stage mainly involved:

- Verifying image-label pairs
- Organizing data into training and testing folders
- Validating the dataset structure
- Configuring dataset paths using the data.yaml file

This ensured compatibility with the YOLOv8 training pipeline.

In [11]:

```
import os

dataset_path =
"/content/drive/MyDrive/blur_person_project/dataset"

print("Dataset Contents:")
print(os.listdir(dataset_path))
```

```
Dataset Contents:
['README.dataset.txt', 'README.roboflow.txt', 'test',
'train', 'data.yaml']
```

```
In [12]: train_images =
len(os.listdir("/content/drive/MyDrive/blur_person_project/
dataset/train/images"))
test_images =
len(os.listdir("/content/drive/MyDrive/blur_person_project/
dataset/test/images"))

print("Training Images:", train_images)
print("Testing Images:", test_images)
```

```
Training Images: 364
Testing Images: 91
```

```
In [13]: with
open("/content/drive/MyDrive/blur_person_project/dataset/data.yaml", "r") as f:
    print(f.read())
```

```
train: ../train/images
val: ../test/images
test: ../test/images

nc: 1
names: ['person']
```

Model Training

YOLOv8n pretrained on the COCO dataset was selected as the baseline model.

The model was fine-tuned using the blurred person detection dataset with customized hyperparameters.

```
In [ ]: from ultralytics import YOLO
import os

def train_model():
    model = YOLO("yolov8n.pt")

    model.train(
        data="dataset/data.yaml",
        epochs=150,
        imgsz=640,
        batch=16,
        lr0=0.001,
        weight_decay=0.0005,
        patience=15,
        multi_scale=True,
        project="results",
        name="blur_finetune",
        save=True,
        verbose=True
    )

train_model()
```

In [21]:

```
# =====  
# Model Evaluation  
# =====  
  
from ultralytics import YOLO  
import os  
  
# Check if required files exist  
model_path =  
"/content/drive/MyDrive/results/blur_finetune_pretrained/weights/best.pt"  
data_yaml =  
"/content/drive/MyDrive/blur_person_project/dataset/data.yaml"  
  
print("Model Exists:", os.path.exists(model_path))  
print("Data YAML Exists:", os.path.exists(data_yaml))  
  
# Load trained model  
print("\nLoading trained model...")  
model = YOLO(model_path)  
  
# Run validation on test dataset  
print("Running validation on test dataset...\n")  
  
metrics = model.val(  
    data=data_yaml,  
    split="test",  
    verbose=True  
)  
  
# Display evaluation metrics  
print("\n==== Validation Results ====")  
print(f"Precision    : {metrics.box.mp:.4f}")  
print(f"Recall      : {metrics.box.mr:.4f}")  
print(f"mAP@50      : {metrics.box.map50:.4f}")  
print(f"mAP@50-95   : {metrics.box.map:.4f}")
```

```
Model Exists: True  
Data YAML Exists: True
```

```
Loading trained model...  
Running validation on test dataset...
```

```

Ultralytics 8.4.57 🚀 Python-3.12.13 torch-2.11.0+cpu
CPU (Intel Xeon CPU @ 2.20GHz)
Model summary (fused): 72 layers, 3,005,843
parameters, 0 gradients, 8.1 GFLOPs
[34m[1mval: [0mFast image access ✅ (ping: 0.7±0.3
ms, read: 19.9±4.5 MB/s, size: 37.7 KB)
[K[34m[1mval: [0mScanning
/content/drive/MyDrive/blur_person_project/dataset/tes
t/labels.cache... 91 images, 2 backgrounds, 0 corrupt:
100% ————— 91/91 20.1Mit/s 0.0s
WARNING ⚠️ Box and segment counts should be equal,
but got len(segments) = 832, len(boxes) = 1003. To
resolve this only boxes will be used and all segments
will be removed. To avoid this please supply either a
detect or segment dataset, not a detect-segment mixed
dataset.
[K
Class      Images  Instances
Box(P      R      mAP50  mAP50-95): 100%
————— 6/6 5.0s/it 29.8s
              all      91      1003
0.911      0.898      0.928      0.671
Speed: 10.4ms preprocess, 290.8ms inference, 0.0ms
loss, 1.8ms postprocess per image
Results saved to [1m/content/runs/detect/val-2[0m

```

==== Validation Results ====

```

Precision   : 0.9110
Recall      : 0.8984
mAP@50      : 0.9277
mAP@50-95   : 0.6706

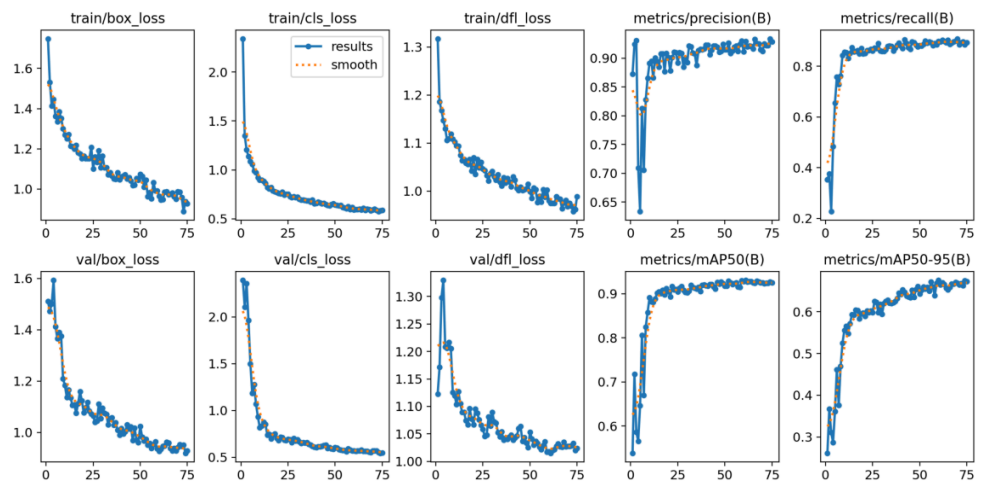
```

```
In [15]: from PIL import Image
import matplotlib.pyplot as plt

img = Image.open(

"/content/drive/MyDrive/results/blur_finetune_pretrained/re
sults.png"
)

plt.figure(figsize=(15,10))
plt.imshow(img)
plt.axis("off")
plt.show()
```



Confusion Matrix

The confusion matrix provides a visual representation of the model's classification performance by comparing predicted labels with actual labels.

A strong diagonal pattern indicates that the model correctly identifies the majority of person instances in the dataset.

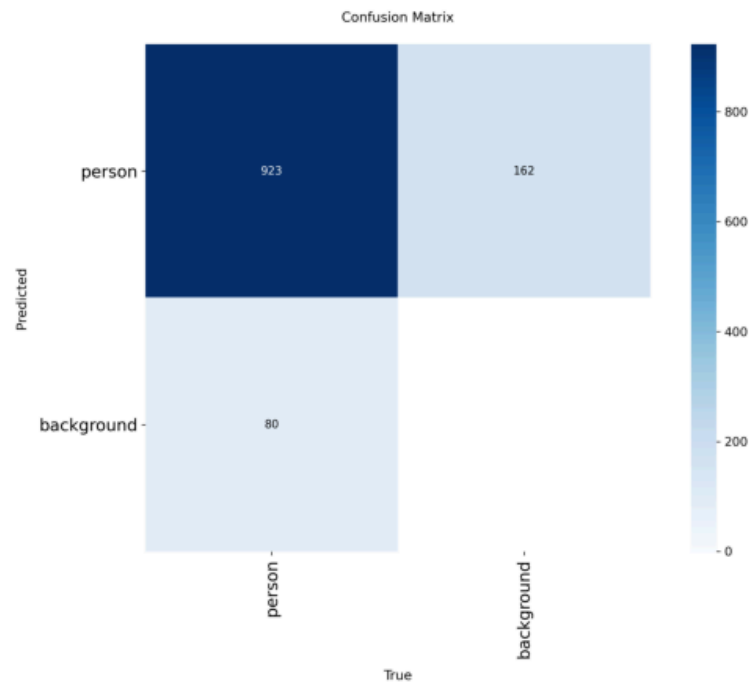
In [16]:

```
from PIL import Image
import matplotlib.pyplot as plt

img = Image.open(

"/content/drive/MyDrive/results/blur_finetune_pretrained/co
nfusion_matrix.png"
)

plt.figure(figsize=(8,8))
plt.imshow(img)
plt.axis("off")
plt.show()
```



Performance Metrics

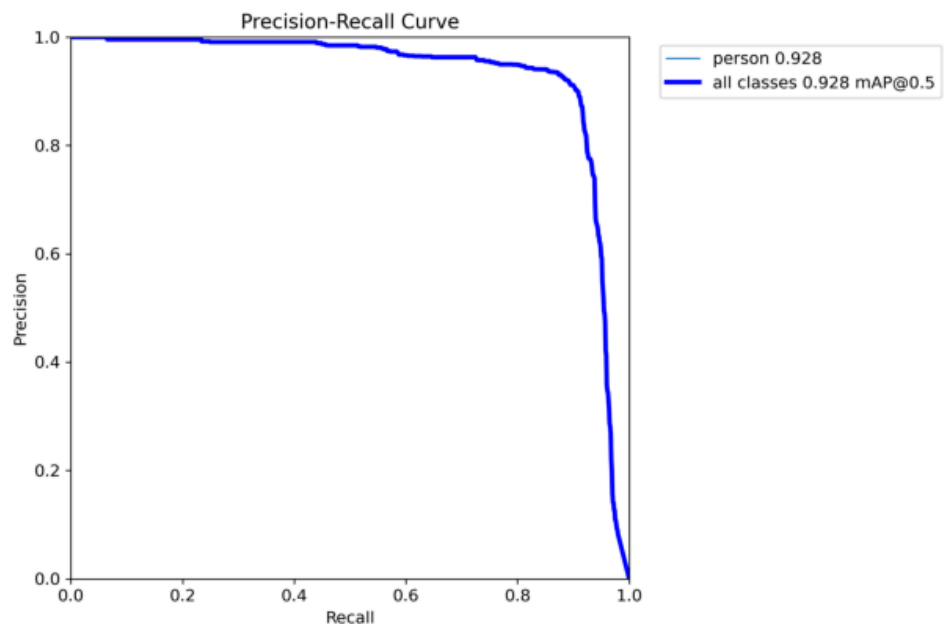
The final model performance was evaluated using standard object detection metrics.

These metrics provide insight into detection accuracy and localization quality on the test dataset.

```
In [23]: img = Image.open(

"/content/drive/MyDrive/results/blur_finetune_pretrained/BoxPR_curve.png"
)

plt.figure(figsize=(8,6))
plt.imshow(img)
plt.axis("off")
plt.show()
```



In [24]:

```
import pandas as pd

df = pd.read_csv(

    "/content/drive/MyDrive/results/blur_finetune_pretrained/re
sults.csv"
)

# Display last training epoch
df.tail(1)
```

Out[24]:

	epoch	time	train/box_loss	train/cls_loss	tra
74	75	734.257	0.92678	0.5864	0.9



```
In [25]: final_row = df.iloc[-1]

metrics_df = pd.DataFrame({
    "Metric": [
        "Precision",
        "Recall",
        "mAP@50",
        "mAP@50-95"
    ],
    "Value": [
        final_row["metrics/precision(B)"],
        final_row["metrics/recall(B)"],
        final_row["metrics/mAP50(B)"],
        final_row["metrics/mAP50-95(B)"]
    ]
})

metrics_df
```

Out[25]:

	Metric	Value
0	Precision	0.92815
1	Recall	0.89332
2	mAP@50	0.92511
3	mAP@50-95	0.67320

Model Prediction

The trained YOLOv8 model was used to perform person detection on unseen blurred test images.

```
In [26]: from ultralytics import YOLO

# Load trained model
model = YOLO(

"/content/drive/MyDrive/results/blur_finetune_pretrained/weights/best.pt"
)

# Run prediction
results = model.predict(

source="/content/drive/MyDrive/blur_person_project/dataset/test/images",
    save=True,
    conf=0.25
)
```

image 1/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame10000_png.rf.7cd20c0ee58250443d033e09c203d411.jpg: 480x640 49 persons, 165.3ms

image 2/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame10005_png.rf.081a53cbc20bf231eb351fe0e6c50e28.jpg: 480x640 52 persons, 156.0ms

image 3/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame10025_png.rf.191e8d63fc906232625b6e7ee38db43c.jpg: 480x640 51 persons, 161.9ms

image 4/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame10080_png.rf.1150c8753067fc9d608130ffcb863fb4.jpg: 480x640 56 persons, 144.6ms

image 5/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame100_png.rf.814313dfb8b85330c1b65371a8aa0126.jpg: 480x640 5 persons, 240.6ms

image 6/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame10135_png.rf.10e9c2d2eddd3592043cdf2a00b8fdf4.jpg: 480x640 44 persons, 232.0ms

image 7/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame1020_png.rf.d9dc296603b05575bdc2cfd9ee1505a2.jpg: 480x640 3 persons, 227.8ms

image 8/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame10285_png.rf.91e3e956a52d47c019140c0b551c0c1a.jpg: 480x640 46 persons, 227.6ms

image 9/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame10290_png.rf.5ee1516e4faaee81defff0ecdf9a5dfc.jpg: 480x640 43 persons, 225.7ms

image 10/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame10320_png.rf.6bc72869c7c5e0b33d0485421bc5d908.jpg: 480x640 46 persons, 215.0ms

image 11/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame10330_png.rf.e899b54ab849aaa6527d83bc984ddd14.jpg: 480x640 41 persons, 222.8ms

image 12/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame1200_png.rf.9aa6e65e148ed83e377d4c0fd79d56dd.jpg: 480x640 4 persons, 231.9ms

image 13/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame1240_png.rf.96b12a26e22aa35976b31522a7624953.jpg: 480x640 5 persons, 224.8ms

image 14/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame1340_png.rf.cfcfae654d4ed3c77f211a91453ba8fc.jpg: 480x640 5 persons, 225.0ms

image 15/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame1600_png.rf.4ce5fce168b8f86108958a7fb220e6e4.jpg: 480x640 6 persons, 224.2ms

image 16/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame160_png.rf.0b0e40eb0f7386f8d81dac343ce2dd56.jpg: 480x640 4 persons, 216.4ms

image 17/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame1640_png.rf.307cf6292fd608116ce2aa348dda24bd.jpg: 480x640 6 persons, 239.7ms

image 18/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame1660_png.rf.92b4459127a7b4abc82830ccf6961832.jpg: 480x640 5 persons, 218.1ms
image 19/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame1680_png.rf.8e97dadbc34fc1493c06f9cd082e84e0.jpg: 480x640 5 persons, 228.9ms
image 20/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame2060_png.rf.5cad6a9783cbf84173a2fb9f619a0c7a.jpg: 480x640 6 persons, 230.0ms
image 21/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame2100_png.rf.2ead83d14a36bcdcb75212ee0cdd7d6b.jpg: 480x640 6 persons, 242.7ms
image 22/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame220_png.rf.f74d376053248058ee2c101231b9d281.jpg: 480x640 4 persons, 226.7ms
image 23/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame2420_png.rf.4f06d34f633d13fddc1fd58839c82f07.jpg: 480x640 7 persons, 227.3ms
image 24/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame2460_png.rf.a3d57dc3b4b313382e97fbde3468f37d.jpg: 480x640 6 persons, 163.7ms
image 25/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame2540_png.rf.51b7cb44d1d6b271026440b885359901.jpg: 480x640 4 persons, 141.4ms
image 26/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame2560_png.rf.ea38c8477852ad0d0d85e9b5381a288e.jpg: 480x640 4 persons, 160.1ms
image 27/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame2640_png.rf.3eb13cfc0c145b760de64f9a7dbaac20.jpg: 480x640 4 persons, 145.0ms
image 28/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame2660_png.rf.5f4d0d56f26d6699b30a02af543eb04b.jpg: 480x640 4 persons, 144.6ms
image 29/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame2860_png.rf.9cfb477d6a790ecaf1d12ea012d20d55.jpg: 480x640 3 persons, 146.7ms
image 30/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame2880_png.rf.a5a0593f8c0582bbf9481ce2b54ee590.jpg: 480x640 3 persons, 145.4ms
image 31/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame3020_png.rf.73dd419fc3f25c18f461edb870ba0bdf.jpg: 480x640 3 persons, 149.3ms
image 32/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame3140_png.rf.4d097fc3bad5031702df520a4d8a4dcd.jpg: 480x640 3 persons, 146.6ms
image 33/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame3180_png.rf.6a64397072616c64fdc9d7c37c3c9e92.jpg: 480x640 3 persons, 149.2ms
image 34/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame3240_png.rf.acbe697c630350d6e155a05d7ce9505d.jpg: 480x640 3 persons, 143.6ms
image 35/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame3340_png.rf.81c7783079a6d98ba3aa19c0e659b704.jpg: 480x640 3 persons, 148.9ms
image 36/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame340_png.rf.56d7ba18125b33a91bd655c43da647f7.jpg: 480x640 4 persons, 146.2ms
image 37/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame3580_png.rf.8b0a8c474a9b216ed457bc510910a060.jpg: 480x640 4 persons, 148.8ms
image 38/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame3780_png.rf.59ba959c2c648aba07a5443dcd9aaf03.jpg: 480x640 1 person, 143.3ms
image 39/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame3840_png.rf.c4ccce1de3268d337694cb27b62e0e98.jpg: 480x640 1 person, 158.5ms
image 40/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame3860_png.rf.86cd7252615e6f8ee54bebdb53b51162.jpg: 480x640 1 person, 143.5ms
image 41/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame3920_png.rf.4f27de16071695ef1633931b2d279a7d.jpg: 480x640 1 person, 144.8ms
image 42/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40024_png.rf.7a6efa0542cf123d51abaf66ba00fb56.jpg: 480x640 13 persons, 144.9ms
image 43/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40032_png.rf.b51c2595b8ff1e779161eb4d1ddcf810.jpg: 480x640 13 persons, 145.3ms
image 44/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40072_png.rf.ad4e56cb3974cc870a742c3709c0aff5.jpg: 480x640 14 persons, 147.2ms
image 45/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40144_png.rf.7d094ff717ff6cef236c6120d8151e99.jpg: 480x640 16 persons, 162.2ms
image 46/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40176_png.rf.d45a5a125cd42c5480ffaeed3951423f.jpg: 480x640 19 persons, 141.6ms
image 47/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40184_png.rf.80396cc6add95a0ffd1c24d421bb8dfb.jpg: 480x640 18 persons, 146.8ms
image 48/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame4020_png.rf.11902210507d96a7d38fe5dbbafeda93.jpg: 480x640 1 person, 147.3ms
image 49/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40256_png.rf.355cef84f845bb508c4194152a1181f7.jpg: 480x640 17 persons, 149.7ms
image 50/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40272_png.rf.3a99616b12c96fd65434ecb3c6ab1a14.jpg: 480x640 14 persons, 148.5ms
image 51/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40280_png.rf.dff64c3e21276cf14b9e00dde7075199.jpg: 480x640 18 persons, 146.7ms
image 52/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40288_png.rf.d81e71cf39c22f3f2ecac2e4d6c96df5.jpg: 480x640 17 persons, 159.5ms
image 53/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40296_png.rf.9397957aeb7a5124f8ed8addc977575d.jpg: 480x640 15 persons, 149.7ms
image 54/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40368_png.rf.1792d05450bbb1e094f6b76a8cff2f3e.jpg: 480x640 15 persons, 148.3ms
image 55/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40400_png.rf.c44095c46c2dbdd3a1fdb7ee78b787c.jpg: 480x640 14 persons, 145.2ms
image 56/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40432_png.rf.3eb1e77827e5808b2dd290bde8e472ed.jpg: 480x640 13 persons, 147.1ms
image 57/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40488_png.rf.16c71909165d0c55ce6283206dc03c40.jpg: 480x640 12 persons, 150.3ms
image 58/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40544_png.rf.7f3d39a574c73ce0b91b69d37e15c123.jpg: 480x640 14 persons, 160.1ms
image 59/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40552_png.rf.0548c2e10df59a465e6bf1f606de0f99.jpg: 480x640 15 persons, 145.9ms
image 60/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40592_png.rf.19910535b87b0bc696d386f13681c39b.jpg: 480x640 14 persons, 148.0ms
image 61/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40656_png.rf.aa477ccd5e1b213c8c4601b5e9bb6a28.jpg: 480x640 13 persons, 146.9ms
image 62/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40664_png.rf.f16072ccd595a74d50e11c4e21d14348.jpg: 480x640 13 persons, 154.1ms
image 63/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40680_png.rf.5398ffa4ebb431d4407b28e9f785eb9f.jpg: 480x640 14 persons, 146.1ms
image 64/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40696_png.rf.199a03d7600066ca752dca8fc84eacd9.jpg: 480x640 14 persons, 160.3ms
image 65/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40720_png.rf.9eb3c272b126d1291bd15172edb8d155.jpg: 480x640 15 persons, 145.9ms
image 66/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40736_png.rf.c336f9fac3e69d590200fb3c831d4f0a.jpg: 480x640 13 persons, 151.1ms
image 67/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40832_png.rf.605f0a1f3ced854a9ac66e38474a53bb.jpg: 480x640 15 persons, 152.5ms
image 68/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40840_png.rf.185486eada4dc2925dd8c50b9b3b0310.jpg: 480x640 17 persons, 152.1ms
image 69/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40952_png.rf.3ad0ca908aab4acc73785224ab52074b.jpg: 480x640 13 persons, 147.9ms
image 70/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40960_png.rf.5e0cd910ceebe50b55b170dc97df7173.jpg: 480x640 13 persons, 163.0ms
image 71/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame40976_png.rf.5c18af92906c5d5df158a62d22119e68.jpg: 480x640 13 persons, 146.4ms
image 72/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame4100_png.rf.8792063ba1cb3c6759093b225e91b934.jpg: 480x640 (no detections), 154.9ms
image 73/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame41032_png.rf.6c5ccbe96c83e5d78271b28ebbf
cfff2c.jpg: 480x640 12 persons, 149.7ms
image 74/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame41080_png.rf.a09ea19acbcabfb7f72dcdf7110
41390.jpg: 480x640 15 persons, 151.8ms
image 75/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame41088_png.rf.ecb770ad4854e40cac39cf41455
d2df3.jpg: 480x640 16 persons, 149.8ms
image 76/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame41184_png.rf.8bf937d8445708b0b0800e2819e
cad42.jpg: 480x640 13 persons, 155.9ms
image 77/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame41200_png.rf.5c7abe37d6038d08499f46f7e97
8c36d.jpg: 480x640 12 persons, 145.4ms
image 78/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame41208_png.rf.7a5008a9876a4fa425cf48e444f
0723d.jpg: 480x640 12 persons, 169.6ms
image 79/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame4120_png.rf.5767de643fe8b6cd1c598e59bf70
606c.jpg: 480x640 (no detections), 151.0ms
image 80/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame41256_png.rf.dd19d0d9ce6074dab865e32c06a
5436a.jpg: 480x640 12 persons, 144.8ms
image 81/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame4180_png.rf.9d11fae0c521a9e4c38cfe5e30b0
af12.jpg: 480x640 1 person, 149.8ms
image 82/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame4200_png.rf.e97cf1177ed2cb33c60a40353db8
23c5.jpg: 480x640 3 persons, 148.3ms
image 83/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame4220_png.rf.2461a9b6a54f18a5b6aef5035ee9
8dd6.jpg: 480x640 3 persons, 165.9ms
image 84/91

/content/drive/MyDrive/blur_person_project/dataset/test/images/frame4280_png.rf.7e0453ae96ffa66f0b362b37ddff480b.jpg: 480x640 3 persons, 165.8ms
image 85/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame4300_png.rf.dc2cd184ed3dc1a4afeba76e4508d322.jpg: 480x640 4 persons, 233.7ms
image 86/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame4460_png.rf.afec8b44b2be0212b3c3a9e620567345.jpg: 480x640 2 persons, 226.1ms
image 87/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame660_png.rf.8305a69bcff6770cfd9c5837f4103a59.jpg: 480x640 3 persons, 232.8ms
image 88/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame760_png.rf.b60951a6c6ce4068500a7b4d0bffa732.jpg: 480x640 3 persons, 235.1ms
image 89/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame780_png.rf.bd4c222c6cce9afcc55fb01365a4a6f0.jpg: 480x640 3 persons, 240.9ms
image 90/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame800_png.rf.5e04fc2a1cc7a95d5d4a5526297542e2.jpg: 480x640 3 persons, 218.4ms
image 91/91
/content/drive/MyDrive/blur_person_project/dataset/test/images/frame900_png.rf.fbb0392274bce496376eedfab30165b.jpg: 480x640 4 persons, 220.2ms
Speed: 2.4ms preprocess, 172.9ms inference, 1.0ms postprocess per image at shape (1, 3, 480, 640)
Results saved to [1m/content/runs/detect/predict][0m

In [27]:

```
import os
```

```
pred_dir = "/content/runs/detect/predict"
```

```
print(os.listdir(pred_dir))
```

```
['frame2420_png.rf.4f06d34f633d13fddc1fd58839c82f07.jpg',  
'frame4200_png.rf.e97cf1177ed2cb33c60a40353db823c5.jpg',  
'frame780_png.rf.bd4c222c6cce9afcc55fb01365a4a6f0.jpg',  
'frame40832_png.rf.605f0a1f3ced854a9ac66e38474a53bb.jpg',  
'frame40432_png.rf.3eb1e77827e5808b2dd290bde8e472ed.jpg',  
'frame40144_png.rf.7d094ff717ff6cef236c6120d8151e99.jpg',  
'frame40488_png.rf.16c71909165d0c55ce6283206dc03c40.jpg',  
'frame4300_png.rf.dc2cd184ed3dc1a4afeba76e4508d322.jpg',  
'frame4120_png.rf.5767de643fe8b6cd1c598e59bf70606c.jpg',  
'frame1680_png.rf.8e97dadbc34fc1493c06f9cd082e84e0.jpg',  
'frame41200_png.rf.5c7abe37d6038d08499f46f7e978c36d.jpg',  
'frame760_png.rf.b60951a6c6ce4068500a7b4d0bffa732.jpg',  
'frame40184_png.rf.80396cc6add95a0fffd1c24d421bb8dfb.jpg',  
'frame40976_png.rf.5c18af92906c5d5df158a62d22119e68.jpg',  
'frame1240_png.rf.96b12a26e22aa35976b31522a7624953.jpg',  
'frame100_png.rf.814313dfb8b85330c1b65371a8aa0126.jpg',  
'frame40736_png.rf.c336f9fac3e69d590200fb3c831d4f0a.jpg',  
'frame40280_png.rf.dff64c3e21276cf14b9e00dde7075199.jpg',  
'frame40296_png.rf.9397957aeb7a5124f8ed8addc977575d.jpg']
```

'frame4220_png.rf.2461a9b6a54f18a5b6aef5035ee98dd6.jpg',
,
'frame2640_png.rf.3eb13cfc0c145b760de64f9a7dbaac20.jpg',
,
'frame10330_png.rf.e899b54ab849aaa6527d83bc984ddd14.jpg',
,
'frame1020_png.rf.d9dc296603b05575bdc2cfd9ee1505a2.jpg',
,
'frame40720_png.rf.9eb3c272b126d1291bd15172edb8d155.jpg',
,
'frame10025_png.rf.191e8d63fc906232625b6e7ee38db43c.jpg',
,
'frame3920_png.rf.4f27de16071695ef1633931b2d279a7d.jpg',
,
'frame41184_png.rf.8bf937d8445708b0b0800e2819ecad42.jpg',
,
'frame40288_png.rf.d81e71cf39c22f3f2ecac2e4d6c96df5.jpg',
,
'frame40696_png.rf.199a03d7600066ca752dca8fc84eacd9.jpg',
,
'frame2860_png.rf.9cfb477d6a790ecaf1d12ea012d20d55.jpg',
,
'frame2100_png.rf.2ead83d14a36bcdcb75212ee0cdd7d6b.jpg',
,
'frame4180_png.rf.9d11fae0c521a9e4c38cfe5e30b0af12.jpg',
,
'frame10290_png.rf.5ee1516e4faaee81defff0ecdf9a5dfc.jpg',
,
'frame40680_png.rf.5398ffa4ebb431d4407b28e9f785eb9f.jpg',
,
'frame3780_png.rf.59ba959c2c648aba07a5443dcd9aaf03.jpg',
,
'frame41208_png.rf.7a5008a9876a4fa425cf48e444f0723d.jpg',
,
'frame40664_png.rf.f16072ccd595a74d50e11c4e21d14348.jpg',
,
'frame4100_png.rf.8792063ba1cb3c6759093b225e91b934.jpg',
,
'frame900_png.rf.fbb0392274bce496376eedfab30165b.jpg',
,
'frame41080_png.rf.a09ea19acbcabfb7f72dcdf711041390.jpg',
,
'frame1340_png.rf.cfcfae654d4ed3c77f211a91453ba8fc.jpg',
,

'frame2540_png.rf.51b7cb44d1d6b271026440b885359901.jpg'
,
'frame3240_png.rf.acbe697c630350d6e155a05d7ce9505d.jpg'
,
'frame4020_png.rf.11902210507d96a7d38fe5dbbafeda93.jpg'
,
'frame40176_png.rf.d45a5a125cd42c5480ffaeed3951423f.jpg',
'frame2560_png.rf.ea38c8477852ad0d0d85e9b5381a288e.jpg'
,
'frame41256_png.rf.dd19d0d9ce6074dab865e32c06a5436a.jpg',
'frame40272_png.rf.3a99616b12c96fd65434ecb3c6ab1a14.jpg',
'frame40840_png.rf.185486eada4dc2925dd8c50b9b3b0310.jpg',
'frame2880_png.rf.a5a0593f8c0582bbf9481ce2b54ee590.jpg',
'frame1660_png.rf.92b4459127a7b4abc82830ccf6961832.jpg',
'frame1200_png.rf.9aa6e65e148ed83e377d4c0fd79d56dd.jpg',
'frame40544_png.rf.7f3d39a574c73ce0b91b69d37e15c123.jpg',
'frame800_png.rf.5e04fc2a1cc7a95d5d4a5526297542e2.jpg',
,
'frame40368_png.rf.1792d05450bbb1e094f6b76a8cff2f3e.jpg',
'frame10320_png.rf.6bc72869c7c5e0b33d0485421bc5d908.jpg',
'frame10285_png.rf.91e3e956a52d47c019140c0b551c0c1a.jpg',
'frame40032_png.rf.b51c2595b8ff1e779161eb4d1ddcf810.jpg',
'frame660_png.rf.8305a69bcff6770cfd9c5837f4103a59.jpg',
,
'frame340_png.rf.56d7ba18125b33a91bd655c43da647f7.jpg',
,
'frame3020_png.rf.73dd419fc3f25c18f461edb870ba0bdf.jpg',
,
'frame40592_png.rf.19910535b87b0bc696d386f13681c39b.jpg',
'frame220_png.rf.f74d376053248058ee2c101231b9d281.jpg',
,

'frame3840_png.rf.c4ccec1de3268d337694cb27b62e0e98.jpg'
,
'frame160_png.rf.0b0e40eb0f7386f8d81dac343ce2dd56.jpg'
,
'frame3140_png.rf.4d097fc3bad5031702df520a4d8a4dcd.jpg'
,
'frame1640_png.rf.307cf6292fd608116ce2aa348dda24bd.jpg'
,
'frame3180_png.rf.6a64397072616c64fdc9d7c37c3c9e92.jpg'
,
'frame2660_png.rf.5f4d0d56f26d6699b30a02af543eb04b.jpg'
,
'frame41088_png.rf.ecb770ad4854e40cac39cf41455d2df3.jpg',
'frame40072_png.rf.ad4e56cb3974cc870a742c3709c0aff5.jpg',
'frame3580_png.rf.8b0a8c474a9b216ed457bc510910a060.jpg',
'frame40952_png.rf.3ad0ca908aab4acc73785224ab52074b.jpg',
'frame41032_png.rf.6c5ccbe96c83e5d78271b28ebbfccff2c.jpg',
'frame40656_png.rf.aa477ccd5e1b213c8c4601b5e9bb6a28.jpg',
'frame10080_png.rf.1150c8753067fc9d608130ffcb863fb4.jpg',
'frame10135_png.rf.10e9c2d2eddd3592043cdf2a00b8fdf4.jpg',
'frame40960_png.rf.5e0cd910ceebe50b55b170dc97df7173.jpg',
'frame1600_png.rf.4ce5fce168b8f86108958a7fb220e6e4.jpg',
,
'frame10000_png.rf.7cd20c0ee58250443d033e09c203d411.jpg',
'frame4280_png.rf.7e0453ae96ffa66f0b362b37ddff480b.jpg',
,
'frame3340_png.rf.81c7783079a6d98ba3aa19c0e659b704.jpg',
,
'frame2060_png.rf.5cad6a9783cbf84173a2fb9f619a0c7a.jpg',
,
'frame40552_png.rf.0548c2e10df59a465e6bf1f606de0f99.jpg',
'frame10005_png.rf.081a53cbc20bf231eb351fe0e6c50e28.jpg',
,

```
'frame2460_png.rf.a3d57dc3b4b313382e97fbde3468f37d.jpg',  
,  
'frame4460_png.rf.afec8b44b2be0212b3c3a9e620567345.jpg',  
,  
'frame40256_png.rf.355cef84f845bb508c4194152a1181f7.jpg',  
,  
'frame3860_png.rf.86cd7252615e6f8ee54bebdb53b51162.jpg',  
,  
'frame40400_png.rf.c44095c46c2dbdd3a1fdb7ee78b787c.jpg',  
,  
'frame40024_png.rf.7a6efa0542cf123d51abaf66ba00fb56.jpg']
```

```
In [28]: from PIL import Image  
import matplotlib.pyplot as plt  
  
img = Image.open(  
    "/content/runs/detect/predict/" +  
    os.listdir("/content/runs/detect/predict")[0]  
)  
  
plt.figure(figsize=(10,8))  
plt.imshow(img)  
plt.axis("off")  
plt.show()
```



Comparison of Detection Results

To evaluate the impact of fine-tuning, predictions from the original COCO-pretrained YOLOv8 model and the blur-finetuned YOLOv8 model were compared on the same blurred image.

This comparison helps visualize the improvement achieved through domain-specific training.

```
In [38]: import os

test_path =
"/content/drive/MyDrive/blur_person_project/dataset/test/im
ages"

image_name = os.listdir(test_path)[0]
image_path =
"/content/drive/MyDrive/blur_person_project/dataset/test/im
ages/frame10285_png.rf.91e3e956a52d47c019140c0b551c0c1a.jpg
"
```

```
In [46]: from ultralytics import YOLO
import matplotlib.pyplot as plt

# Load models
blur_model = YOLO(

"/content/drive/MyDrive/results/blur_finetune_pretrained/weights/best.pt"
)

coco_model = YOLO("yolov8n.pt")

blur_results = blur_model.predict(image_path, conf=0.25,
verbose=False)
coco_results = coco_model.predict(image_path, conf=0.25,
verbose=False)

blur_count = len(blur_results[0].boxes)
coco_count = len(coco_results[0].boxes)

# Draw detections
blur_img = blur_results[0].plot()
coco_img = coco_results[0].plot()

# Show side-by-side
fig, ax = plt.subplots(1, 2, figsize=(16, 8))

ax[0].imshow(blur_img)
ax[0].set_title(
    f"Blur-Finetuned YOLOv8\nDetected Persons:
{blur_count}",
    fontsize=12
)
ax[0].axis("off")

ax[1].imshow(coco_img)
ax[1].set_title(
    f"COCO Pretrained YOLOv8\nDetected Persons:
{coco_count}",
    fontsize=12
)
ax[1].axis("off")
```

```
plt.tight_layout()
plt.show()
```

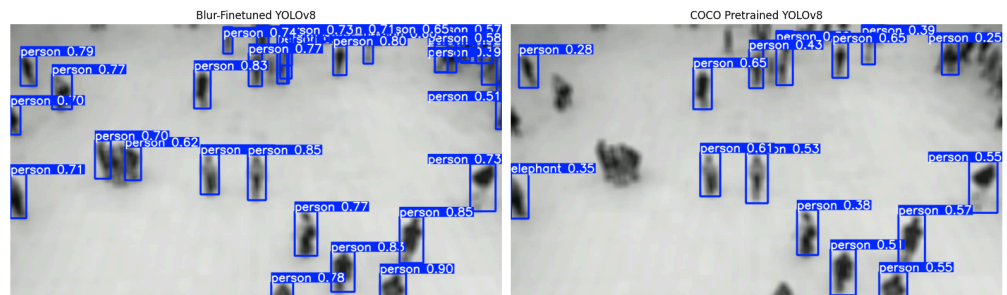
```
print(f"Blur-Finetuned YOLOv8 Detections : {blur_count}")
print(f"COCO Pretrained YOLOv8 Detections: {coco_count}")
```

0: 384x640 37 persons, 195.2ms

Speed: 6.4ms preprocess, 195.2ms inference, 2.1ms
postprocess per image at shape (1, 3, 384, 640)

0: 384x640 14 persons, 1 elephant, 135.2ms

Speed: 3.4ms preprocess, 135.2ms inference, 1.3ms
postprocess per image at shape (1, 3, 384, 640)



Blur-Finetuned detections: 37

COCO detections: 15

Conclusion

In this project, a YOLOv8n model was fine-tuned on a custom blurred-person dataset to improve person detection performance in degraded visual environments.

The trained model achieved:

- Precision: 0.911
- Recall: 0.898
- mAP@50: 0.928
- mAP@50-95: 0.671

Comparative experiments showed that the blur-finetuned model detected significantly more persons in blurred scenes than the original COCO-pretrained model. This demonstrates the effectiveness of domain-specific fine-tuning for robust object detection.

Future work may include expanding the dataset, testing larger YOLO models, and evaluating performance under different weather and lighting conditions.